7.7 Quantities That Never Change

WORKSHEET

The question

STEP III 2024, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself The expression $\mathbf{MN} - \mathbf{NM}$ is called the *commutator* of $\mathbf{M}$ and $\mathbf{N}$. When is it zero? What is its trace, given part (i)?

STAGE 2 The derivative of a determinant

Ask yourself Differentiate $\det \mathbf{M} = ad - bc$ by the product rule. Then compute $\mathbf{M}^{-1}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}$ and its trace, using $\mathbf{M}^{-1} = \frac{1}{\det \mathbf{M}}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

STAGE 3 Three conserved quantities

Ask yourself Take the trace of $\dot{\mathbf{M}} = \mathbf{MN} - \mathbf{NM}$. For the determinant, use part (ii): what is $\mathbf{M}^{-1}(\mathbf{MN} - \mathbf{NM})$?

STAGE 4 An explicit solution

Pause and try With $\mathbf{N} = t\mathbf{I} + t\mathbf{E}$, where $\mathbf{E} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, compute $\mathbf{MN} - \mathbf{NM}$ and write down the differential equations for a, b, c, d. Solve them in the order c, then a and d, then b.

STAGE 5 A different equation

Ask yourself If $\dot{\mathbf{M}} = \mathbf{MN}$, then $\mathbf{N} = \mathbf{M}^{-1}\dot{\mathbf{M}}$, and by part (ii) $\operatorname{tr}\mathbf{N}$ is the logarithmic derivative of $\det \mathbf{M}$. Can the trace of $\mathbf{M}$ stay constant while its determinant changes?

Try this next

1. **Check the solution.** Verify by differentiating that the matrix $\mathbf{M}(t)$ in Stage 4 satisfies $\dot{\mathbf{M}} = \mathbf{MN} - \mathbf{NM}$.

2. **Similarity.** Show that if $\mathbf{M}(t) = \mathbf{S}^{-1}\mathbf{M}_0\mathbf{S}$, where $\mathbf{M}_0$ is constant and $\mathbf{S}(t)$ satisfies $\dot{\mathbf{S}} = \mathbf{SN}$, then $\dot{\mathbf{M}} = \mathbf{MN} - \mathbf{NM}$. (*Hint:* $\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{S}^{-1} = -\mathbf{S}^{-1}\dot{\mathbf{S}}\mathbf{S}^{-1}$.)

3 **Volume growth.** Let $\mathbf{M}(t) = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{-t} \end{pmatrix}$. Find $\mathbf{N}$ with $\dot{\mathbf{M}} = \mathbf{M}\mathbf{N}$, and check that $\frac{\mathrm{d}}{\mathrm{d}t} \ln \det \mathbf{M} = \operatorname{tr} \mathbf{N}$.

4 **Higher powers.** Show directly that $\frac{\mathrm{d}}{\mathrm{d}t} \operatorname{tr}(\mathbf{M}^k) = 0$ for every positive integer k when $\dot{\mathbf{M}} = \mathbf{M}\mathbf{N} - \mathbf{N}\mathbf{M}$.

Further reading

- **P. G. Drazin & R. S. Johnson, *Solitons: An Introduction*** (Cambridge University Press). The standard undergraduate introduction, including Lax pairs.
- **Thierry Dauxois & Michel Peyrard, *Physics of Solitons*** (Cambridge University Press). Includes the story of Scott Russell's wave.
- **Dwight Neuenschwander, *Emmy Noether's Wonderful Theorem*** (Johns Hopkins University Press). How symmetries give conservation laws.

The full guided solution, and what the question leads to at university, is in Section 7.7:
https://beyondstep.keithmcnulty.org/ch07/lax-pairs.html